

Course Code: CIT 316

Lab 1 - Worksheet 1: Immutable Infrastructure & Containerization

Objective: To transform raw application code into a versioned, portable Docker image and manage its lifecycle via the command line.

PHASE 1: THE BUILD CONTEXT

TASK 1.1

Establish a local directory and create the source artifact (index.html). Note that in DevOps, the folder containing your Dockerfile is known as the **Build Context**.

```
mkdir devops-lab && cd devops-lab  
echo "<h1>DevOps Lab v1.0</h1>" > index.html
```

PHASE 2: DEFINING THE IMAGE (THE DOCKERFILE)

A Dockerfile is a declarative script that defines the environment. Unlike manual server setup, this allows for 100% reproducibility.

```
# Use Nginx (Alpine Linux version) as our base  
FROM nginx:alpine  
  
# Inject our code into the standard Nginx path  
COPY index.html /usr/share/nginx/html/index.html  
  
# Documentation for networking  
EXPOSE 80
```

PHASE 3: EXECUTION & VERSIONING

TASK 3.1

Build the image and tag it. Tags allow DevOps engineers to manage versions (e.g., v1, v2, latest).

```
docker build -t my-first-web-app:v1 .
```

TASK 3.2

Instantiate the container. We use **Port Mapping** to link the internal container port to our host machine.

```
docker run -d -p 8080:80 --name web-v1 my-first-web-app:v1
```

Supplementary Tech-Savvy Q&A

Scenario	Engineering Explanation
Why did my build fail initially?	Docker requires a file named exactly "Dockerfile" without an extension. Common errors include "Dockerfile.txt" (hidden extension).
What is Port 8080:80?	8080 is the <i>Host Port</i> (your laptop). 80 is the <i>Container Port</i> (where Nginx listens). This prevents port conflicts on a shared server.
Is the container a VM?	No. A VM includes a full OS. A container shares the Host's Kernel, making it much lighter and faster to boot (milliseconds vs minutes).

PHASE 4: THE UPDATE LIFECYCLE

In a CI/CD pipeline, we never "patch" a running container. We destroy the old one and deploy a new one. This is called **Immutable Infrastructure**.

1. Modify `index.html` to "Version 2.0".
2. Run `docker build -t my-first-web-app:v2 .`
3. Remove the old container: `docker rm -f web-v1`
4. Run the new version: `docker run -d -p 8080:80 --name web-v2 my-first-web-app:v2`

End of Worksheet 1 - Proceed to Lab 2: Kubernetes Orchestration